



DATA ANALYST PORTFOLIO PROJECT

Restaurant Success Analysis

What the Yelp dataset reveals about user engagement, ratings,
and the formula behind restaurant success

Yelp Open Dataset

SQL · Python · Statistical Testing · NLP Sentiment · Time Series

THE DATASET

Five Tables, One Yelp Ecosystem

The Yelp Open Dataset covers businesses across 8 North American metro areas. Restaurants are isolated from the broader business table before any analysis begins.

150K+

Total businesses

8

Metro areas (US & Canada)



business

Name, location, categories, star rating, review count, open status



review

Star rating, date, full text, and useful / funny / cool votes



tip

Short tips left by users, with date and business reference



checkin

Comma-separated string of all check-in timestamps per business



user

User profiles, including review count and Elite status

WHY THIS MATTERS

Defining Success Beyond a Star Rating



Problem Statement

In a competitive market like the restaurant industry, understanding what drives business success is crucial for stakeholders. This project uses the Yelp dataset to investigate the relationship between user engagement — reviews, tips, and check-ins — and business success metrics such as review count and average star rating.

Research Objectives

1

Quantify the correlation between user engagement (reviews, tips, check-ins) and review count / average star rating.

2

Analyse the impact of review sentiment on restaurant ratings, using NLP scoring on actual review text.

3

Explore whether consistent engagement over time is associated with sustained business success.

HYPOTHESES

Three Bets Going Into the Analysis

H1



Engagement drives metrics

Higher levels of user engagement (more reviews, tips, and check-ins) correlate with higher review counts and ratings.

H2



Sentiment drives rating

Positive sentiment in review text is associated with higher star ratings.

H3



Consistency drives success

Successful restaurants (high ratings) exhibit consistent and growing user engagement over time.

Verdicts for each hypothesis are revealed as the evidence builds throughout this deck.

METHODOLOGY

Cleaning a Heavily Skewed Dataset

Out of 150K+ businesses, 35,004 are open restaurants. Their review counts are heavily right-skewed — a handful of chains would dominate any aggregate analysis — so outliers are removed using the IQR method before proceeding.

Before Outlier Removal		After Outlier Removal	
Restaurants	35,004	Restaurants	31,537
Avg. review count	104.1	Avg. review count	56.0
Median review count	15	Median review count	15
Max review count	7,568	Max review count	248
Avg. star rating	3.52	Avg. star rating	3.48



Why it matters

The median (15) sits far below the mean (104) before cleaning — a classic right-skew. After trimming with the IQR rule, the average review count drops from 104 to 56 and the maximum from 7,568 to 248, giving a far more representative picture of a typical restaurant.

EXPLORATORY ANALYSIS

Popular Isn't the Same as Well-Loved



Most Reviewed (Chains)

Restaurant	Reviews	Rating
McDonald's	16,490	1.9
Chipotle Mexican Grill	9,071	2.4
Taco Bell	8,017	2.1
Chick-fil-A	7,687	3.4
First Watch	6,761	3.9
Panera Bread	6,613	2.7



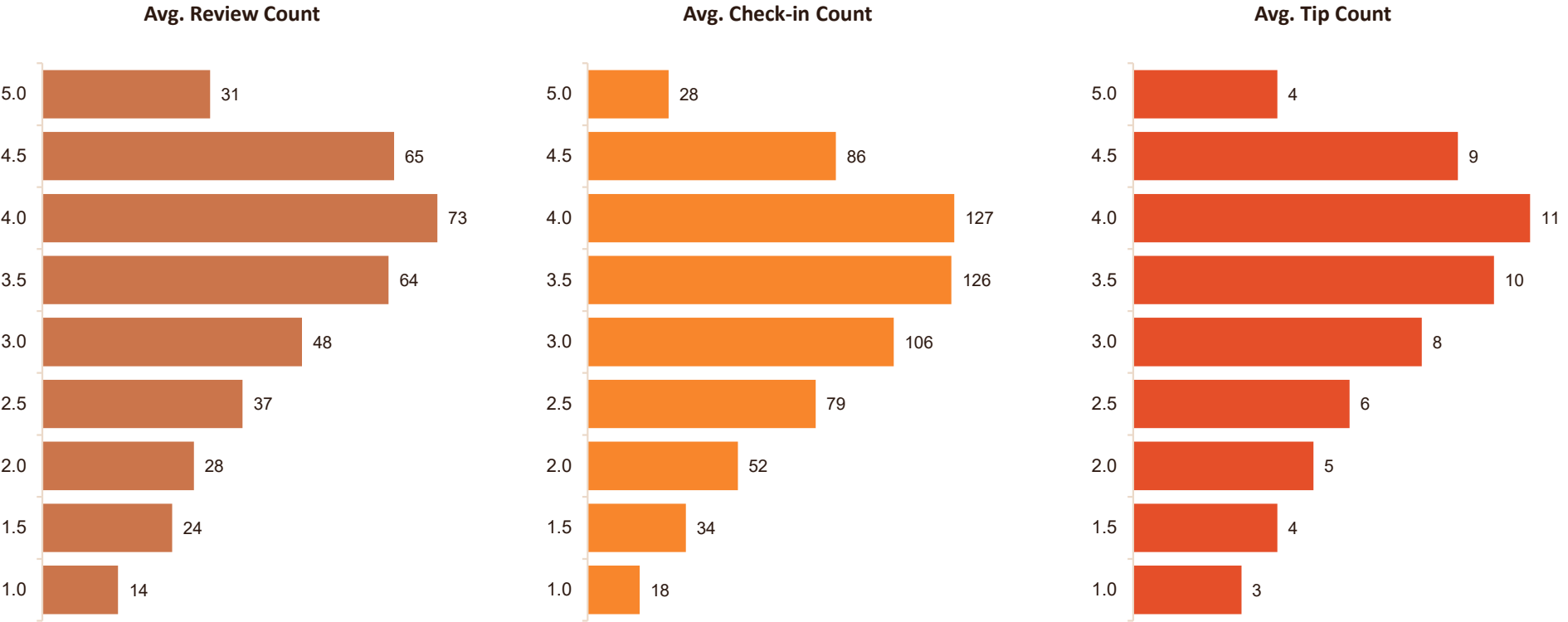
Highest Rated (Independents)

Restaurant	Reviews	Rating
ā café	48	5.0
two birds cafe	77	5.0
la bamba	44	5.0
taqueria la cañada	17	5.0
la 5th av tacos	24	5.0
cabbage vegetarian cafe	12	5.0

Insight: Review count and star rating capture different dimensions of presence. A restaurant can be popular without being well-regarded, and vice versa — which is why neither metric alone defines “success.”

DOES ENGAGEMENT TRANSLATE TO HIGHER RATINGS?

Engagement Peaks at 4 Stars, Then Falls



All three engagement signals rise steadily from 1 ★ to 4 ★, then drop at 4.5 ★–5 ★ — a ceiling effect among small, loyal audiences rather than a linear relationship.

STATISTICAL VALIDATION

Engagement Metrics Align — Rating Lags Behind

Engagement vs. Rating

	rating	review	checkin	tip
rating	1.00	0.68	0.43	0.57
review	0.68	1.00	0.91	0.98
checkin	0.43	0.91	1.00	0.97
tip	0.57	0.98	0.97	1.00

Between Engagement Channels

	review	checkin	tip
review	1.00	0.70	0.75
checkin	0.70	1.00	0.75
tip	0.75	0.75	1.00

Significance Testing (Pearson & Spearman vs. Rating)

Metric	Pearson r	Spearman ρ	Sig.
Review count	0.245	0.231	Yes
Check-in count	0.079	0.051	Yes
Tip count	0.163	0.140	Yes

p < 0.05 for all three metrics — statistically significant, but Pearson r of 0.05–0.25 means the relationships are weak in practical terms.



H1 Verdict: Partially Supported

Engagement and rating move together, and high-rated restaurants attract significantly more engagement overall (Mann-Whitney, $p < 0.05$). But engagement volume alone explains very little of the variation in rating — popularity and quality are related, not the same thing.

A COMPOSITE METRIC

Building a Success Score, City by City

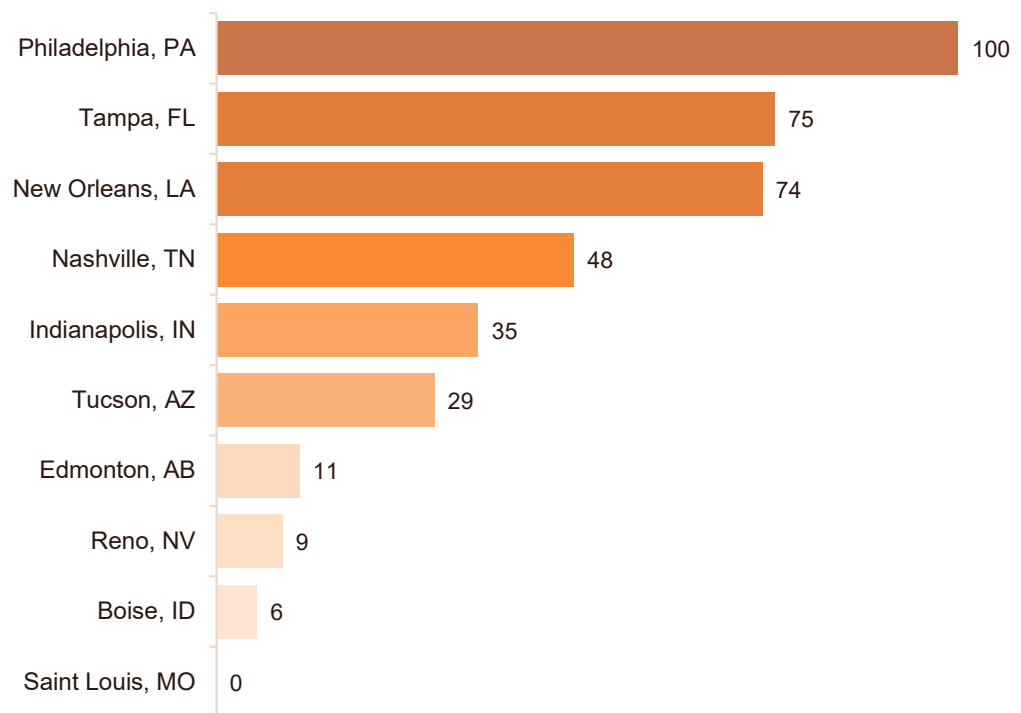


The Formula

$$\text{rating} \times \log(\text{reviews} + 1)$$

Neither rating nor review count alone captures success, so the two are combined. The logarithm dampens the effect of very high review counts, so a restaurant with 10,000 reviews doesn't completely dominate one with 500. The result is normalised to a 0–100 scale.

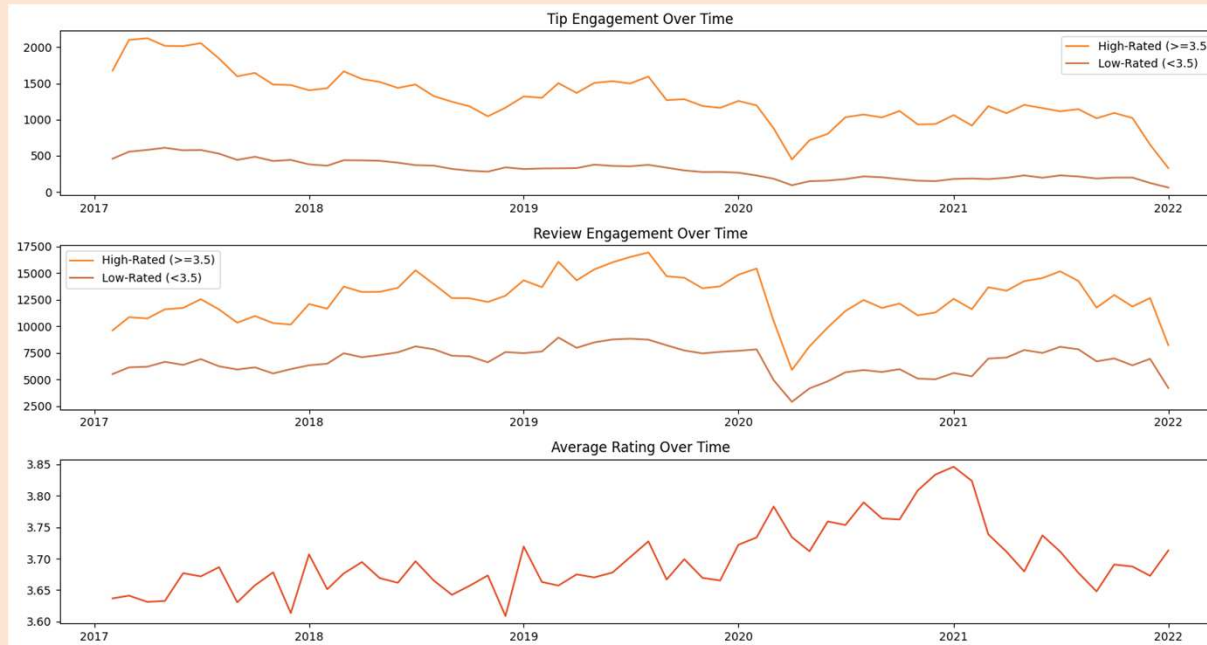
Success Score by Metro Area (Top 10)



Philadelphia leads, combining strong average ratings with high review volume — a useful benchmark market for restaurant best practices.

LONGITUDINAL ANALYSIS

High-Rated Restaurants Sustain Engagement



Monthly review & tip volume, High-Rated (≥ 3.5 ★) vs. Low-Rated restaurants, 2017–2022



High-rated restaurants consistently out-pace low-rated ones in both tip and review volume across the full five-year window.



A sharp 2020 dip and rebound reflects pandemic disruption — engagement recovers faster for high-rated restaurants.



Seasonal decomposition confirms a recurring Nov–Mar peak each year, likely tied to holiday dining.

H3 Verdict: Supported

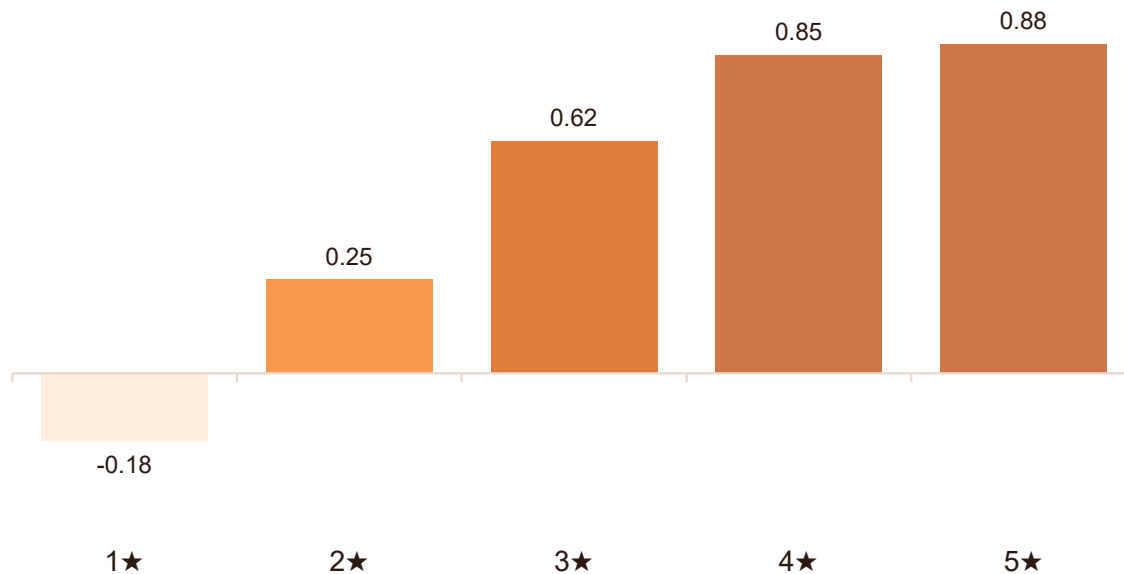
Consistently growing engagement over time is a defining feature of high-rated restaurants.

WHAT DO THE REVIEWS ACTUALLY SAY?

Review Sentiment Tracks Star Rating Closely

Beyond engagement volume, what about the content of reviews? Using VADER — a lexicon-based sentiment scorer built for short social text — 10,000 sampled reviews were scored from -1 (most negative) to +1 (most positive).

Mean VADER Sentiment Score by Star Rating



$r = 0.64$

Pearson correlation between
VADER score and star rating

85%

of sampled reviews scored as
Positive sentiment overall

H2 Verdict: Strongly Supported

Sentiment rises monotonically from 1★ to 5★ and correlates far more strongly with rating than engagement volume did — what reviewers write aligns closely with what they rate.

WHO'S BEHIND THE NUMBERS?

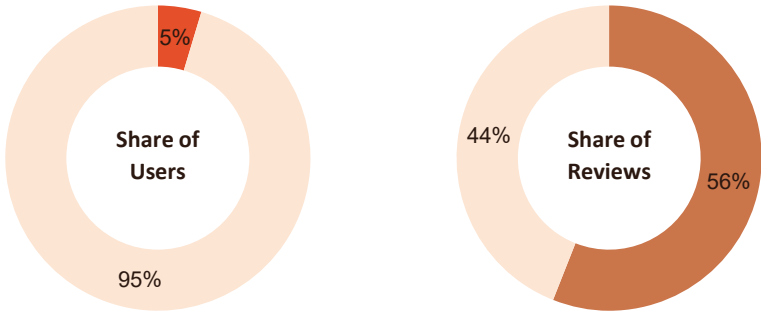
Reactions, Votes & the Elite Minority

Review Votes vs. Success Metrics

	reviews	useful	funny	cool	success
reviews	1.00	0.80	0.64	0.65	0.70
useful	0.80	1.00	0.75	0.80	0.64
funny	0.64	0.75	1.00	0.72	0.45
cool	0.65	0.80	0.72	1.00	0.66
success	0.70	0.64	0.45	0.66	1.00

Useful and Cool votes track the success score more closely than Funny votes do — resonance matters more than humor.

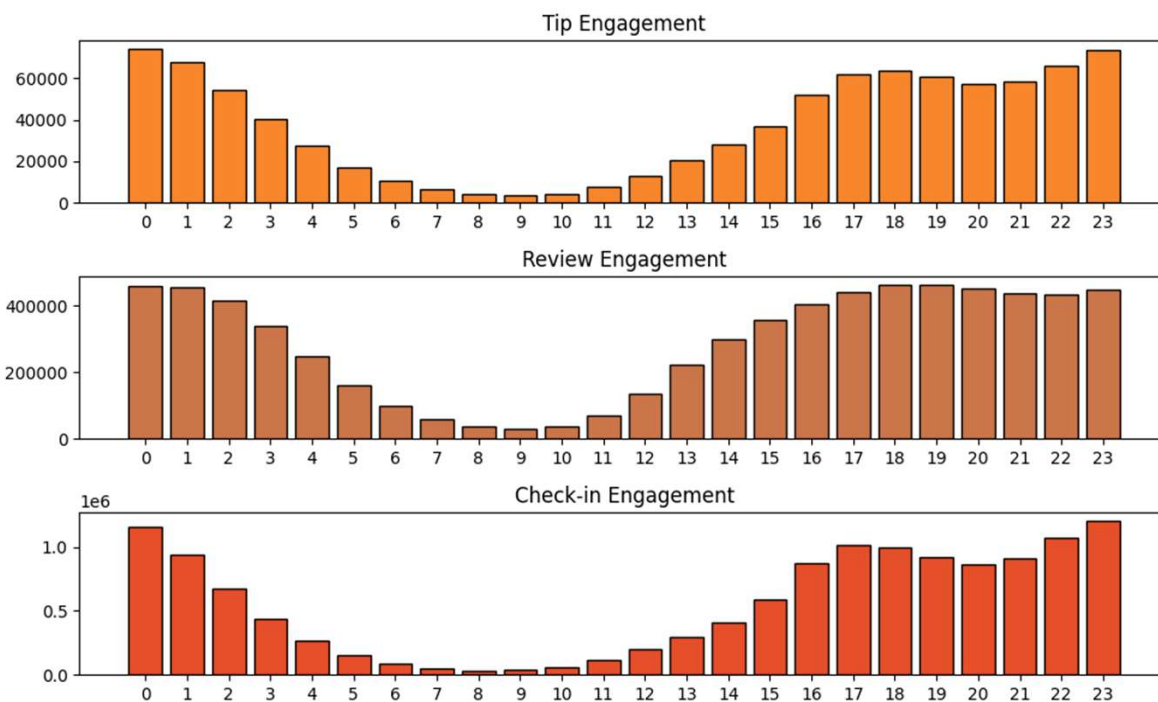
Elite vs. Non-Elite Users



Elite users make up just 4.6% of the user base but contribute 56% of total review volume — a disproportionate share of what shapes a restaurant's public perception.

TIMING INSIGHT

When Do People Actually Engage?



Tip, review, and check-in volume by hour of day (0–23h), across all restaurants

4PM–1AM

the window that dominates reviews, tips, and check-ins alike

Activity concentrates in the evening across all three engagement types — peaking as people arrive, dine, and review afterward. The late-night window captures bars and late-dining spots too.

TURNING INSIGHT INTO ACTION

Recommendations



Chase quality, not just volume

Engagement metrics correlate weakly with rating — focus on experiences that generate positive reviews, not just more of them.



Monitor sentiment, not just stars

VADER scores track ratings closely ($r \approx 0.64$) and shift before aggregate ratings do — an early warning signal.



Target the Nov–Mar window

Seasonal analysis shows this is consistently the highest-engagement period for promotions and launches.



Engage Elite users deliberately

A small group writes a disproportionate share of reviews — getting them in the door has outsized impact.



Benchmark against Philadelphia

It leads the composite success score, combining rating quality with engagement volume.



Staff around peak hours

The 4PM–1AM window dominates engagement — align staffing and service quality to that window.

Caveat: all findings are correlational. A longitudinal or causal-inference design would be needed to confirm the direction of effect.



Thank You

Questions about the methodology, the SQL, or the dataset? Let's talk.
